**7** **(a)** Draw the circuit symbol for a potential divider.

[1]

**(b)** Fig. 7.1 shows a circuit.

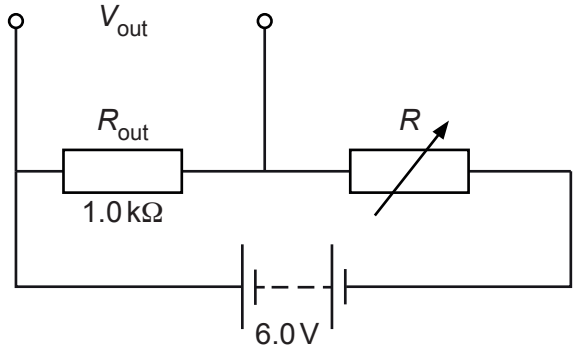



**Fig. 7.1**

**(i)** Calculate the value of $V_{out}$ when the value of $R$ is 3.0 kΩ.

$V_{out}$ = ..... [2]

**(ii)** The value of $R$ is adjusted until the current in the circuit is 1.7 mA.

Calculate the charge that flows through the circuit in 300 s.

charge = ..... [2]

[Total: 5]